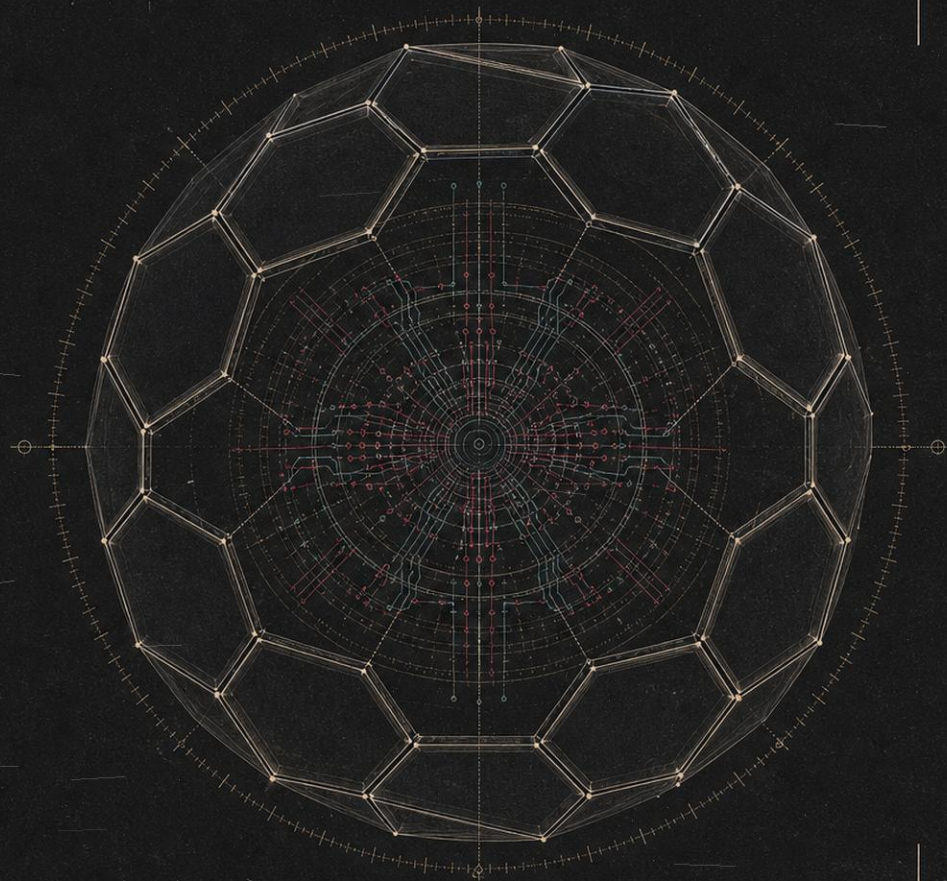


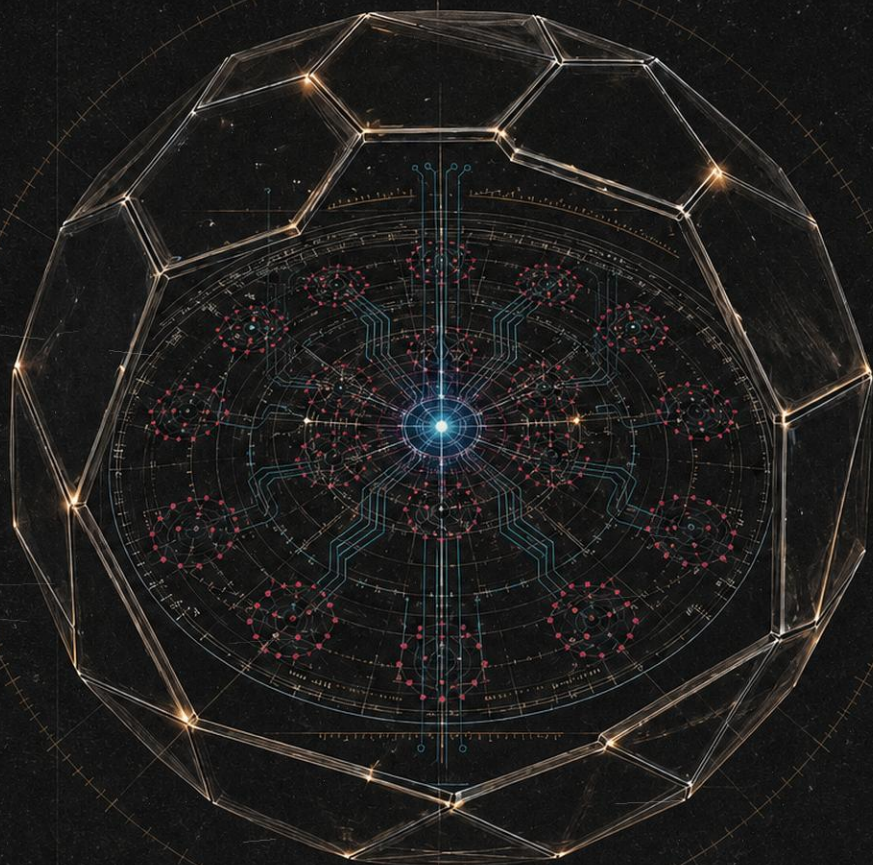
INTRODUCING

AEGIS

VOLUME II

Memory Utility Labs
Calgary, Alberta





a Memory Utility Publication

Introducing

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SYSTEM ARCHITECTURE, VOL. II

Product 1.2.0 · Issue 1 · 2026

Volume I stated the law. This volume states the machinery.

Control plane. Token capacity basin. Guard middleware. Hermes routing.

Honesty yield: savings_percent is billed USD on matched_provider_pairs (D-040). Routing is tiny-chat only. Implement packs stay off.

Absolute Form: austere · aggressive on drift · protective reserve ($\geq 80\%$).

GOLDEN GATE BOOK



CONTENTS

This volume

i	Half title	5	III. Guard middleware
ii	Title	6	allow deny require-review
iii	Contents	7	IV. Hermes routing
1	I. Control plane	8	tiny-chat on; implement packs off
2	kernel / process / frozen /v1	9	Colophon
3	II. Token capacity basin		
4	pack / scrub / reserve as low-flow		

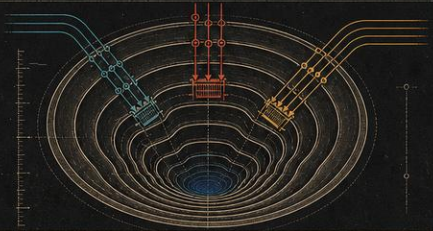
THIS VOLUME

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1 CONTROL PLANE

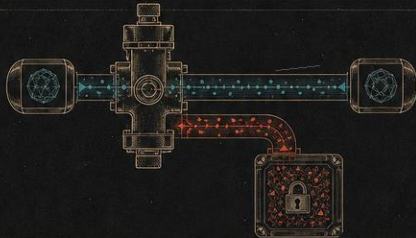


2 TOKEN CAPACITY BASIN

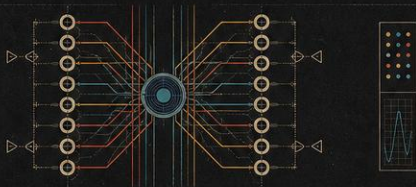


3 GUARD MIDDLEWARE

→ THROUGH-PATH
→ DIVERT PATH



4 HERMES ROUTING SYSTEM



CHAPTER I

Control plane

Kernel (process / memory / drivers / syscalls). Portable data plane under ~/.aegis/. Frozen /v1 API. Honest yield proof.

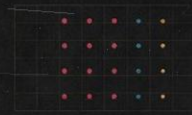
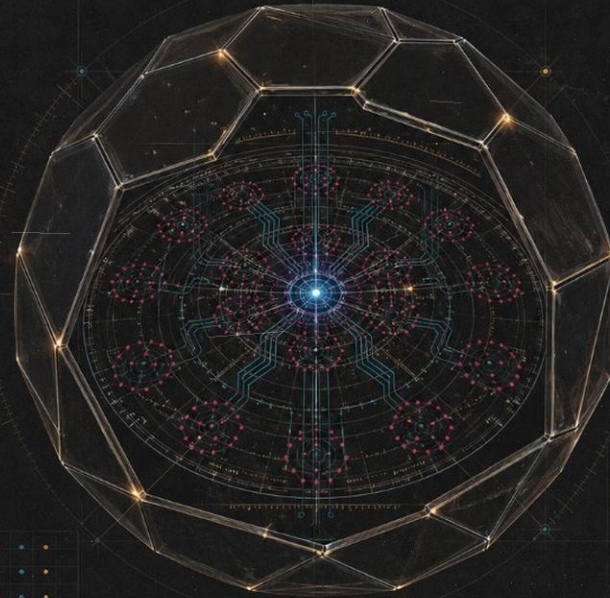
aegis os init lays the plane. aegis doctor --product scores it. aegis os ready is the floor. Optional: aegis serve at <http://127.0.0.1:8787> for OpenAI-compatible calls with body.aegis.pipeline=true.

The serve daemon is not a host kernel. Agent-kernel scores are not Darwin scores. The plane is local. Hosted SaaS, consumer marketplace, and host-kernel replacement are out of scope.

Volume I named the operating system. This chapter is the control surface: process table, frozen contract, local daemon. Nothing here authorizes a remote brain.

AEGIS CONTROL PLANE

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SYSTEM ARCHITECTURE, VOL. I

CHAPTER II

Token capacity basin

Tokens are water. They are not free. Pack and scrub before discharge. A reserve is a low-flow state, not an empty one.

Weekly cap: 1 000 000 processed tokens. Reserve floor: 80%. OPEN / THROTTLE / HARD STOP. Surplus is not decoration: 20% of new savings to the wish jar after audit.

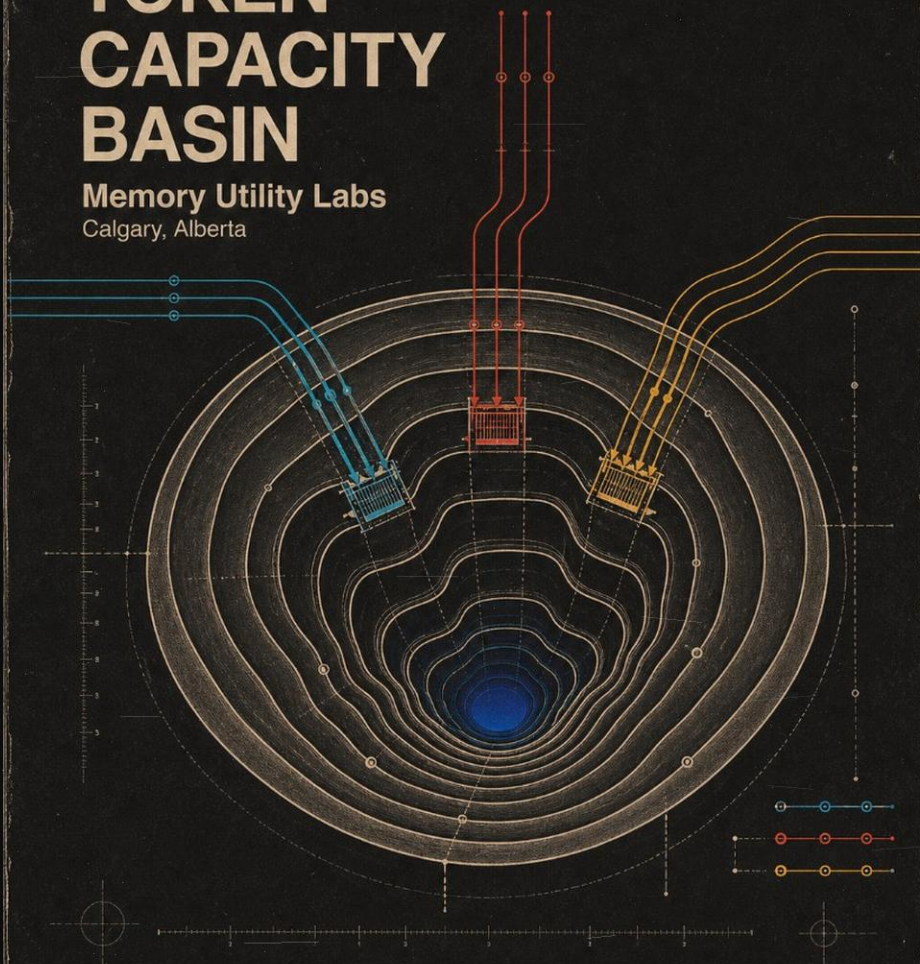
Naïve: dump -> overflow -> re-read. Aegis: pack + reserve + implement-full + land + audit.

The work-system is one basin. Chat windows are jurisdictions, not units. Covering reuse hits if hashes still match. After you edit, the cache misses on purpose.

TOKEN CAPACITY BASIN

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RESOURCE STEWARDSHIP, VOL. III

CHAPTER III

Guard middleware

Single path: request -> normalize -> ids -> classify -> policy -> fail-closed -> allow | deny | require-review -> execute only if permitted -> GuardDecision audit.

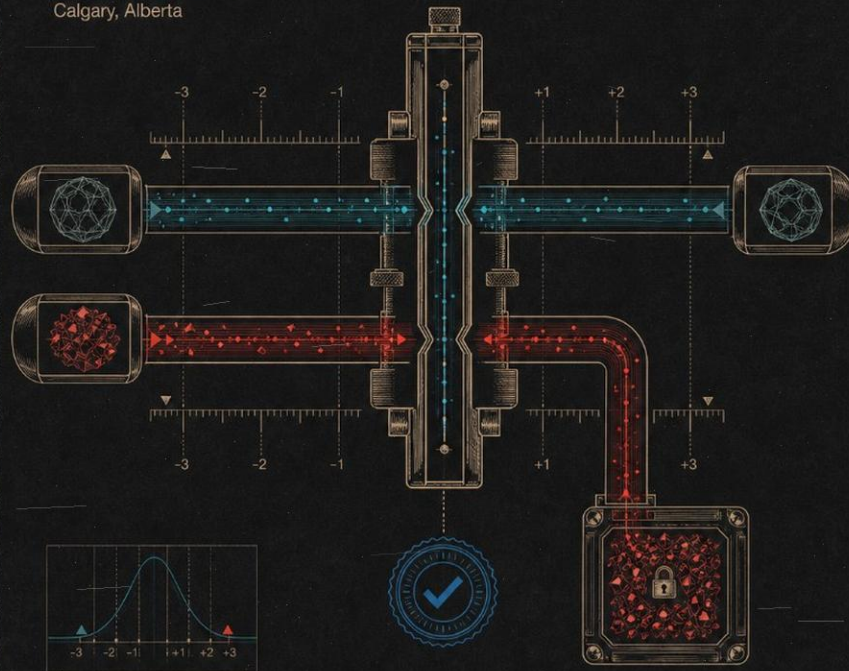
Unknown tools deny. Malformed requests deny. Out-of-scope requests deny. Shadow mode defaults on: the decision is logged; the block is not yet the law. Turn shadow off when the catalog is trusted.

Ledger: `~/aegis/guard_log.jsonl`. Rotate: `aegis guard rotate`. Hermes wrapper never raises to block a tool; it returns a structured deny payload instead.

Hermes middleware itself is fail-open on raise. Guard must not be. Admission is a valve, not a suggestion.

AEGIS GUARD MIDDLEWARE

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ADMISSION & SAFETY, VOL. V

CHAPTER IV

Hermes routing

aegis hermes search|resolve is a thin read-only CLI over the local index. Track first. Perplexity only for a live external miss. Perplexity never edits.

Coordination pathways are drawn. Tiny-chat routing is on (D-040): explore/review v4-pro may swap to nano. Implement packs stay on the requested model. savings_percent is observed billed USD, not a pack guess.

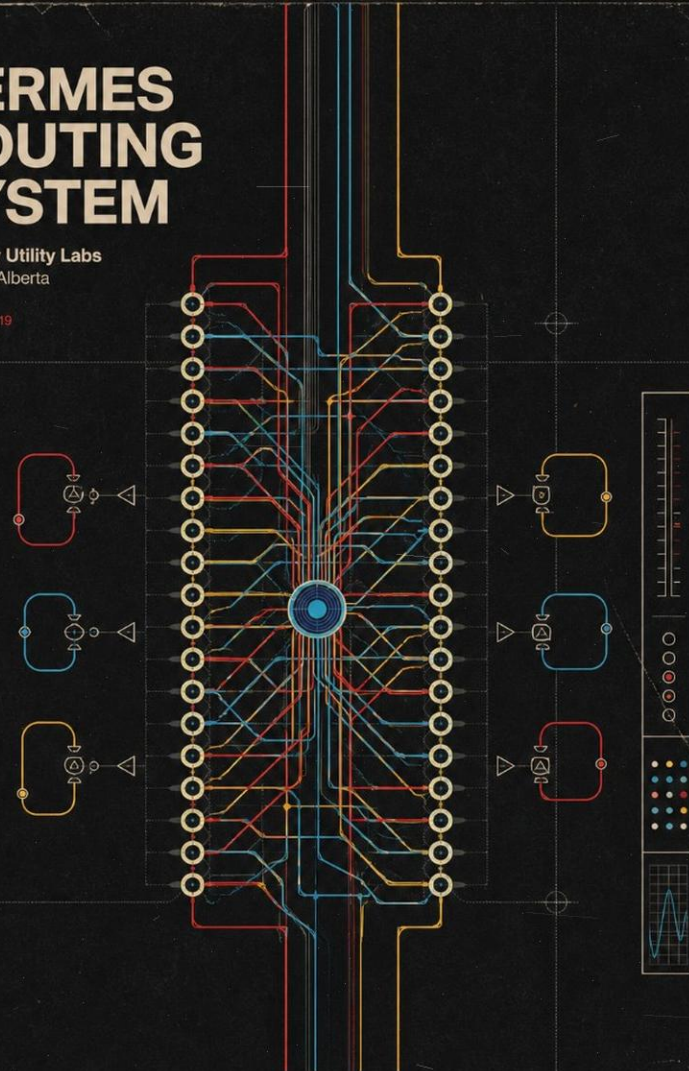
Advertised tool-call metadata is not admission. A research note is not a routing trial. The index is a marshal. It is not a switch.

The index is a marshal. Tiny-chat is a scoped trial, not a global switch. Implement packs are not routed.

HERMES ROUTING SYSTEM

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COORDINATION PATHWAYS, VOL. II

COLOPHON

Memory Utility Labs

Designed as a Golden Gate Book for Memory Utility Labs, Calgary, Alberta.

Volume II plates follow 1970s NASA / GSFC technical-poster language: charcoal ground, cream Helvetica, muted teal / brick / mustard, drafting marks, analog grain. Chapter figures are the four system posters.

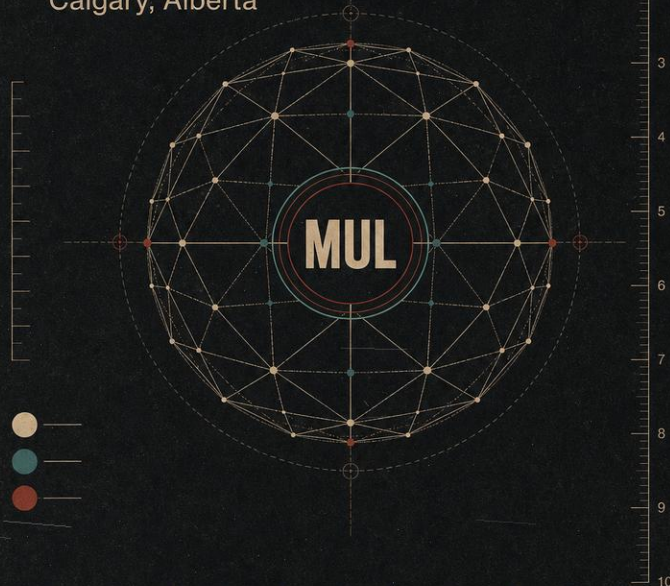
Type: Helvetica. Format: 9 x 6 in landscape, 200 dpi. Product 1.2.0. Issue 1. 2026.

Manuscript from FIELD.md, FIRST_RELEASE.md, D-040. savings_percent is billed USD on matched_provider_pairs.
Tiny-chat routing on. Implement packs unrouted.

Remain in Absolute Form unless explicitly released.

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